LEVEL UP ACADEMY

Class 8 | CBSE / GSEB | Science

Quick Revision: Combustion and Flame

Key ideas

- **Combustion** is a chemical process in which a substance reacts with oxygen to give off heat (and often light).
- Three things are needed: **fuel, air (oxygen), and heating to the ignition temperature**. Remove any one and the fire goes out.
- **Ignition temperature:** the lowest temperature at which a substance catches fire. Inflammable substances (petrol, LPG, alcohol) have very low ignition temperatures.

Type	Example
Rapid combustion	LPG burning on a gas stove
Spontaneous combustion	White phosphorus catching fire in air at room temperature
Explosion	Firecrackers, when heat, light and sound are released suddenly

A candle flame

Zone	What happens	Heat
Outermost (blue, non-luminous)	Complete combustion	Hottest. Goldsmiths use it to melt gold and silver.
Middle (yellow, luminous)	Partial combustion	Moderately hot
Innermost (dark)	Unburnt wax vapour	Least hot

Fire control and fuels

- Water cools the fuel below its ignition temperature, but must **not** be used on electrical fires or oil and petrol fires.
- **Carbon dioxide** is the best extinguisher for electrical and oil fires: it is heavier than oxygen, so it covers the fire like a blanket, and it also cools it.
- A good fuel is cheap, easily available, burns at a moderate rate, has a high calorific value and leaves little residue.
- Calorific value (kJ/kg): hydrogen about 1,50,000 · LPG about 55,000 · petrol and kerosene about 45,000 · coal 25,000–33,000 · wood 17,000–22,000 · cow dung cake 6,000–8,000.
- Harmful effects: carbon monoxide is poisonous; more CO_2 adds to global warming; oxides of sulphur and nitrogen cause acid rain.

Quick check

1. Why does a matchstick catch fire when rubbed on the side of the box, but not on its own?

2. Why is water not used to put out a fire caused by petrol?